RAPTOR Experimental Code Structure and End-to-End Workflow

Shared implementation and protocol feeding three reproducible aerodynamic reconstruction pipelines

DATA & PROTOCOL

AirfRANS 2D

Official Reynolds / AoA tasks
180 / 20 / 200 split
4,096 nodes; 7 → 4 variables
configs/*.yaml

Automobile surfaces

100 vehicles; 80 / 10 / 10
4,096 sampled surface nodes
single-channel surface field
--archive PATH_TO_ARCHIVE

Aircraft surfaces

100 stratified aircraft
J-20 / P180 / B737 / ERJ
80 / 10 / 10; 4,096 nodes
--archive PATH_TO_ARCHIVE

SHARED CORE

Unified representations

```
data.py
training-only normalization
deterministic sampling + split
64-D latent projection
64 physics tokens; 12-NN
```

Models and optimization

```
raptor/model.py
transolver_tiny/ · point baselines
modern Tiny baseline wrappers
AdamW + AMP + clipping
seed 42 + early stopping
```

Verification boundary

```
tests/ · CUDA smoke tests
serial GPU execution
result.json resume markers
no CPU fallback
no overwrite of completed runs
```

EXPERIMENT PIPELINES

AirfRANS extrapolation

```
raptor/run.py
run_aoa_pipeline.py
run_core_ablation_pipeline.py
Reynolds + AoA + 4 ablations
```

100-car matched benchmark

```
drivaernetpp_3d/
run_formal_100car.py
PointNet / Transolver / RAPTOR
paired + regional analysis
```

100-aircraft transfer

```
aircraft_3d/
run_formal_100aircraft.py
Transolver-Tiny vs RAPTOR
family + best/worst analysis
```

EVIDENCE & FIGURES

2D evidence

```
outputs/**/result.json
four-variable errors + resources
scripts/plot_airfrans_*.py
figs/airfrans_*/
```

Automobile evidence

```
per-vehicle MSE + paired tests
geometry-region contributions
full-mesh field/error figures
analysis_consolidated_*/
```

Aircraft evidence

```
per-aircraft + family statistics
four-family overview
best/worst 2 × 6 comparison
analysis/*.csv · *.json · *.pdf
```

Common reproducibility contract

Identical split · identical node count · identical seed · matched width/depth · validation selection · compact JSON/CSV evidence · publication PDF figures